NEXASTORE

Computing for Developers, Creators & AI Builders



GPU SELECTION

GPU Selection Guide for AI, Gaming and Creators

One GPU, three very different workload profiles

NexaStore Knowledge Series • 2026 Edition

1. Select the GPU for the Workload

GPU selection changes depending on whether the buyer values local AI, gaming frame rate, creative acceleration or a combination. VRAM is critical for many AI and creator workloads, while gaming also depends heavily on raster performance, resolution, refresh rate and laptop power limits.

Buyer profile	GPU priority	VRAM direction	Other priorities
AI developer	Runtime compatibility + VRAM	12-16 GB+	RAM, SSD, cooling
Competitive gamer	High frame rate / latency	8-12 GB often sufficient	CPU, high-refresh display
AAA gamer	Raster/RT performance	12-16 GB	QHD display, cooling
Video creator	Codec/effects acceleration	12-16+ GB	CPU, RAM, fast storage
3D / VFX	Compute + VRAM	16-24+ GB	RAM, cooling, expansion
AI + creator	VRAM + software ecosystem	16-24+ GB	64 GB+ RAM, storage

2. Why VRAM Is Workload-Specific

Use	VRAM pressure comes from
Local LLM	Model weights, KV cache, context and concurrency
Computer vision	Resolution, batch size, model architecture and training
Gaming	Textures, resolution, ray tracing and game engine assets
Video / motion	Effects, frame buffers, AI features and timeline complexity
3D	Scene geometry, textures, render buffers and GPU render engines

3. Laptop vs Desktop GPU

Factor	Laptop GPU	Desktop GPU
Mobility	Excellent	None
Power / cooling	Restricted by chassis	Higher sustained headroom
VRAM ceiling	Usually lower	Broader high-VRAM options
Upgradeability	Usually fixed	Often replaceable
Best fit	Mobile gaming/creator/AI	Sustained high-end work

4. Mixed-Use Recommendations

Scenario	Recommendation direction
Gaming + occasional AI	Strong gaming GPU with at least moderate VRAM; avoid overbuying solely for AI.
Developer + local LLM + gaming	Prioritize 12-16+ GB VRAM, 32-64 GB RAM and strong cooling.
Creator + GenAI	Prioritize 16 GB+ VRAM, 64 GB RAM and fast 2 TB+ storage.
3D + AI research	High-VRAM desktop/workstation GPU may provide better value than a thin laptop.
Cloud-first AI + gaming	Buy primarily for gaming; cloud handles heavy AI compute.

5. Recommendation Agent Rules

• Use structured catalog data for exact GPU model, VRAM, power, price and stock.

• Use RAG guides to explain why a given GPU tier fits the workload.

• Do not rank GPUs using VRAM or TOPS alone.

• Ask whether AI workloads are local or cloud-based before recommending premium AI hardware.

• For laptops, include thermals, weight, battery and display in the final recommendation.

Educational notice

This document is part of the fictional NexaStore educational case study.